

Assignment /

Assignment #: 4

Topic : Tuples and Dictionaries

Level 1: Easy

1.TUPLE OPERATIONS

Given an ordered sequence of element with different datatype like integer, list, string(etc..) and perform tuple operation of concatenation.

Input format

Initializing the first tuple t1 with datatype of integer, float, string, list.

Initializing the second tuple t2 with datatype of integer, float, string, list.

Output format

Print the concatenated tuple.

Constraints

Tuple t1 not equal to character datatype.

Tuple t2 not equal to character datatype

NOTE:

Only the same datatypes can be combined with concatenation, an error arises if a list and a tuple are combined.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	The original tuple 1 : (30,9,7) The original tuple 2 : (4, 6)	The original tuple 1 : (8,6,4) The original tuple 2 : (1,9)	The original tuple 1 : (99,76,89) The original tuple 2 : (45,27)	The original tuple 1 : hi The original tuple 2 : ko
Output	The tuple after concatenation is : (30,9,7, 4, 6)	The tuple after concatenation is : (8,6,4,1,9)	The tuple after concatenation is : (99,76,89,45,27)	The tuple after concatenation is : ('h', 'i', 'k', 'o')

2.TUPLE METHODS AND BULID IN FUNCTIONS

Many functions to work on tuples. Reversing a Tuple using the reversed() built-in function.

In this method, we do not make any copy of the tuple. Instead, we get a reverse iterator which we use to cycle through the tuple, similar to the list.

Input format

A tuple test case can have any number of items and they may be of different types (integer, float, list, string, etc.)

Output format

Print the reversed tuple

Constraints

$1 \leq \text{tuple} \leq 10^5$

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	tuples = ('z','a','d','f','g','e','e','k')	tuples = (10, 11, 12, 13, 14, 15)	tuples = ("s","g","l","y","p")	tuples = (33,22,44,55,66,77)
Output	('k', 'e', 'e', 'g', 'f', 'd', 'a', 'z')	(15, 14, 13, 12, 11, 10)	('p', 'y', 'l', 'g', 's')	(77, 66, 55, 44, 22, 33)

3. CREATING A TUPLE WITH MIXED DATATYPES AND NESTED TUPLE.

The sequence of values stored in a tuple can be of any type, and they are indexed by integers.

Input format

Tuples can contain any number of elements and of any datatype (like strings, integers, list, etc.)

Output format

Print tuple with mixed datatype and nested tuple.

Constraints

$1 \leq \text{my tuple} \leq 10^5$

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	my_tuple = () my_tuple = (1, 2, 3) my_tuple =	my_tuple = () my_tuple = (4,5,6)	my_tuple = () my_tuple = (1, 2, 3)	my_tuple = () my_tuple = (1, 2, 3)

	(1, "Hello", 3.4) my_tuple = ("mouse", [8, 4, 6], (1, 2, 3))	my_tuple = (78, "Hi python", [3,5,6]) my_tuple = ("monitor", [8, 4, 6], (89,76,56))	my_tuple = (1, "Hello", 3.4) my_tuple = ("mouse", [8, 4, 6], (1, 2, 3))	my_tuple = (1, "Hello", 3.4) my_tuple = ("mouse", [8, 4, 6], (1, 2, 3))
Output	() (1, 2, 3) (1, 'Hello', 3.4) (('mouse', [8, 4, 6], (1, 2, 3)))	() (1, 2, 3) (78, 'Hi python', [3,5,6]) (('monitor', [8, 4, 6], (89,76,56)))	() (1, 2, 3) (1, 'Hello', 3.4) (('mouse', [8, 4, 6], (1, 2, 3)))	() (1, 2, 3) (1, 'Hello', 3.4) (('mouse', [8, 4, 6], (1, 2, 3)))

4. ADDING ELEMENTS TO A DICTIONARY

pooja is wanted to adding elements in dictionary (may contain duplicates or may be distinct) of dict[index]. pooja has also adding nested elements in its.so can you help her to achieve this task?

Input format

The first line of each test case consists of Dict[value] to adding elements.

The second line of each test case consists of nested elements to adding on dictionary.

Output format

Print the added elements in Dictionary.

Constraints

1 <= N <= 100

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Dict[0] = 'good' Dict[2] = 'For' Dict[3] = 1 Dict[2] = 'Welcome' Dict[5] = {'Nested': {'1': 'Life', '2': 'girl'}}}	Dict[0] = 'apple' Dict[2] = 'good' Dict[3] = 'health' Dict[2] = 'Welcome' Dict[5] = {'Nested': {'1': 'Life', '2': 'easy'}}}	Dict[0] = 'apple' Dict[2] = 'ago' Dict[3] = '123' Dict[2] = 'pleasure' Dict[5] = {'Nested': {'1': 'Life', '2': 'easy'}}}	Dict[0] = '33' Dict[2] = '88' Dict[3] = '123' Dict[2] = 'pleasure' Dict[5] = {'Nested': {'1': '44', '2': '36'}}}
Output	Empty Dictionary: {} Dictionary after adding elements: {0: 'good', 2: 'For', 3: 1} Updated key value: {0: 'good', 2: 'Welcome' Adding a Nested Key: {0: 'good', 2: 'Welcome' {'Nested': {'1': 'Life', '2':	Empty Dictionary: {} Dictionary after adding elements: {0: 'apple', 2: 'g 'health'} Updated key value: {0: 'apple', 2: 'Wel 'health'} Adding a Nested Key: {0: 'apple', 2: 'Wel 'health', 5: {'Neste 'Life', '2': 'easy'}}}	Empty Dictionary: {} Dictionary after adding elements: {0: 'apple', 2: 'g '123'} Updated key value: {0: 'apple', 2: 'plea '123'} Adding a Nested Key: {0: 'apple', 2: 'plea '123', 5: {'Neste 'Life', '2': 'easy'}}}	Empty Dictionary: {} Dictionary after adding elements: {0: '33', 2: '88', 3: '1 Updated key value: {0: '33', 2: 'pleas '123'} Adding a Nested Key: {0: '33', 2: 'pleas '123', 5: {'Nested': '2': '36'}}}

5. ACCESSING ELEMENTS OF A DICTIONARY

In order to access the items of a dictionary refer to its key name. Key can be used inside square brackets.

Input format

The input of dictionary consists of element and key value.

Output format

Print the element accessed by key value.

Constraints

1 <= Dict values <= 10⁵

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
------------	--------------	--------------	--------------	--------------

Input	Dict = {1: 'name', 2: 'address', 3: 'contact number'}	Dict = {1: 'PYTHON', 'name': 'IS', 3: 'EASY'}	Dict = {1: 'CREATE', 'name': 'OWN', 3: 'TOOL'}	Dict = {1: 'GREAT', 'name': 'OWN', 3: 'IDEA'}
Output	Accessing a element using key 2: address Accessing a element using key 1: name	Accessing a element using key name: IS Accessing a element using key 1: PYTHON	Accessing a element using key name: OWN Accessing a element using key 1 : CREATE	Accessing a element using key name: OWN Accessing a element using key 1: GREAT

Level 2: Medium

1. FIND LARGEST ITEM IN A TUPLE

Using Recursive Functions find the largest item in a tuple with integer datatype.

Input format

The first line of each test case consists of tuple item with Positive integers.

Output format

Print the largest item in a tuple.

Constraints

$1 \leq \text{tuple item} \leq 10^5$ (integer datatype)

Sample test Cases

Test Cases	Test Case1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Tuple Items = (11, 65, 54, 23, 76, 33, 82, 98)	Tuple Items = (78,98,65,43,999)	Tuple Items = (987,56,456,876,543, 236)	Tuple Items = (7,8,9,19,45,5)
Output	Maximum Item in Tuple = 98	Maximum Item in Tuple = 999	Maximum Item in Tuple = 987	Maximum Item in Tuple = 19

2. BIDIRECTIONAL TUPLE PAIRS

Given an original list of 2x2 matrix value. Use a for loop to iterate over the tuples and their frequencies in the Counter object. add the product of the frequency of the current tuple and the frequency of the reverse tuple

Input format

The first line of each test case consists of original list of 2x2 matrix with Positive integers.

Output format

print the final count of bidirectional pairs.

Constraints

$1 \leq \text{original list} \leq 10^5$

Note: Input tuple must be integer.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	The original list is : [(5, 6), (1, 2), (6, 5), (9, 1), (6, 5), (2, 1)]	The original list is : [(1, 2), (9, 1), (6, 5), (2, 1)]	The original list is : [(3, 6), (6, 5), (9, 8)]	The original list is : [(1, 2), (9, 1), (6, 5), (2, 1), (2, 1)]
Output	Bidirectional pairs count : 3	Bidirectional pairs count : 1	Bidirectional pairs count : 0	Bidirectional pairs count : 2

3. REMOVE TUPLES OF LENGTH K

There are a set of integer value in the test_list and remove tuple of length k

Input format

The first line of each test case consists of test_list with positive integer.

Output format

Print the tuple of length k

Constraints

$1 \leq K \leq 100$

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	test_list = [(4, 5), (4,), (8, 6, 7), (1,), (3, 4, 6, 7)],	test_list = [(4, 5), (4,), (8, 6, 7), (1,), (3, 4, 6, 7)],	[(4, 5), (4,), (8, 6, 7), (1,), (3, 4, 6, 7)]	[(4, 5), (4,), (8, 6, 7), (1,), (3, 4, 6, 7)]

K	3	2	4	1
Output	[(4, 5), (4,), (1,), 7)]	[(4,), (8, 6, 7), (1, 6, 7)]	[(4, 5), (4,), (8, 6, 7)]	[(4, 5), (8, 6, 7), (3, 4, 6, 7)]

4. HANDLING MISSING KEYS IN PYTHON DICTIONARIES

Mohan given an dictionaries element and its was associated with values. Can you help him to handling its missing key value by error handling methods?

Input format

The input of each test case consists of different elements defined by index value.

The second line of each test case consists of values of N as positive integer.

Output format

Use the 'except' block to handle the KeyError exception and print a message 'Not Found'.

Constraints

1 < = N < = 1000

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	defd['a'] = 1 defd['b'] = 2 defd['d'] = 3	defd['a'] = 1 defd['b'] = 2 defd['d'] = 3	defd['a'] = 1 defd['b'] = 2 defd['d'] = 3	defd['a'] = 1 defd['b'] = 2 defd['d'] = 3
N	7	9	5	6
Output	The value associ 'a' is : 1 The value associ 'c' is : Key Not found	The value associated with 'd' is : 3 The value associated with 'o' is found	The value associated with 'b' is : 1 The value associated with 'g' is : Key Not f	The value associ 'a' is : 1 The value associ 'f' is : Key Not fou

5. POSSIBLE WORDS USING GIVEN CHARACTERS

Given a dictionary and a character array, print all valid words that are possible using characters from the array. Traverse list of given strings one by one and convert them into dictionary using Counter(input) method of collections module.

Input format

The first line of each test case consists of input set values as different words.

The second line of each test case consists of charset elements with different character.

Output format

print possible words formed by character

Constraints

$1 \leq \text{input} \leq 10^5$ (string datatype)

Note: Repetitions of character is not allowed.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	input = ["go","bat","me","eat","goal","run"] charset = ['e','o','b','a','m','g','l']	input = ['goo', 'b', 'eat', 'goal', 'boy'] charSet = ['e', 'o', 'b', 'a', 'm', 'g', 'y']	input = ['mat', 'b', 'sat', 'girl', 'boy'] charSet = ['s', 'o', 'b', 'a', 'm', 'g', 't']	input = ['go', 'hen', 'eat', 'goal', 'run'] charSet = ['e', 'o', 'h', 'a', 'm', 't', 'n']
Output	go, me, goal.	me,boy	sat,mat	eat,hen

Level 3: Hard

1. SORT A LIST OF TUPLES BY SECOND ITEM

Rohan is given a list of tuples to sort the tuples by their second item each tuple has an element in the given list and help him to access the second element of each tuple using the nested loops. This task will be performed by the in-place method of sorting?

Input format

The first line of each test case consists of input as list of tuples.

The input tuple consists of first item and second item with any kind of datatype.

Output format

You need to print the sorted list of tuples by second item.

Constraints

$1 \leq \text{tuple element} \leq 10^5$ (any datatype)

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
------------	--------------	--------------	--------------	--------------

Input	[('for', 24), (('Gods', 8), (('creates', 30))]	[('452', 10), ('256', 5), ('100', 20), (('135', 15))]	[('for', 24), ('i (('home', 28), (('house', 5), ('por (('a', 15))]	[('49', 1), ('56', 5), ('10', 20), (('35', 15))]
Output	[('Gods', 8), ('fo (('creates', 30))]	[('256', 5), ('452', 10 15), ('100', 20)]	[('house', 5), ('is', 15), ('portal', 20 24), ('home', 28)]	[('49', 1), (('56', 5), (('35', 15), (('10', 20)]

2.RESTRICT TUPLES BY FREQUENCY OF FIRST ELEMENT'S VALUE

Given a Tuple list, the task is to write a program to restrict the frequency of the 1st element of tuple values to at most K?

Input format

The first line of each test case consists of K Positive integers.

The second line of each test case consists of tuple list element.

Output format

Print the filtered tuple

Constraints

$1 \leq K \leq 100$

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
K	2	3	1	2
Input	[(2, 3), (3, 3), (1, 4), (2, 4), (2, 5), (3, 4), (1, 4), (3, 4), (4, 7)]	[(2, 3), (3, 3), (1, 4), (2, 4), (2, 5), (3, 4), (1, 4), (3, 4), (4, 7)]	[(6, 8), (5, 9), (7, 9), (9, 6), (2, 5), (3, 4), (1, 4), (5, 5), (7, 7)]	[(5, 5), (7, 7), (9, 6), (2, 5), (3, 4), (7, 7), (5, 5), (7, 7)]
Output	[(2, 3), (3, 3), (1, 4), (2, 4), (3, 4), (1, 4), (4, 7)]	[(2, 3), (3, 3), (1 7)]	[(6, 8), (5, 9), (7 6), (2, 5), (3, 4),	[(5, 5), (7, 7), (9 5), (3, 4), (7, 7),

3.K'TH NON-REPEATING CHARACTER

Given a string and a number k, find the k-th non-repeating character in the string. Consider a large input string with lots of characters and a small character set. How to find the character by only doing only one traversal of input string?

Input format

The input test case consists of string with different and a small character set.

The second line of each test case consists of K positive integer.

Output format

Print the K'th non repeating character.

Constraints

$1 \leq K \leq 1000$

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	str= goods for goods	str= nothing is nothing	str= change only change	str=gods gives best
K	3	4	4	3
Output	r	Less than k non-repeating characters in input.	y	i

4. DICTIONARY TO FIND MIRROR CHARACTERS IN A STRING

Henry is given a string and a number K, we need to mirror the characters from the K-th position up to the length of the string in alphabetical order. In mirror operation, we change 'a' to 'z', 'b' to 'y', and so on so can you please help him find mirror character?

Input format

The first line of each test case consists of K positive integer

The input of each test cases consists of string with traverse characters from length K

Given string and change characters into their mirror value using a dictionary.

Output format

Print the mirror character of the given string.

Constraints

$$1 \leq K \leq 10^5$$

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
K	3	6	4	3
Input	paradox	pneumonia	hesitate	python
Output	paizwlc	pneumlmrz	hesrgzgv	pygs1m

5.COMBINING TWO DICTIONARIES

Combine two dictionaries having key of the first dictionary and value of the second dictionary

Note : There are no duplicates in the key

Input format

Given two dictionaries. The task is to merge them in such a way that the resulting dictionary contains the key from the first dictionary and the value from the second dictionary.

Output format

Print the final mapped dictionary, res.

Constraints

Loop through the range of the length of keys1

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	The original dictionary 1 is : {'Gfg': 20, 'is': 36, 'best': 100} The original dictionary 2 is : {'Gfg2': 26, 'is2': 19, 'best2': 70}	The original dictionary 1 is : {'quality': 65, 'ji': 23, 'hash': 98} The original dictionary 2 is : {'quality2': 90, 'is2': 87, 'hash2': 90}	The original dictionary 1 is : {'keys': 20, 'ji': 36, 'best': 100} The original dictionary 2 is : {'keys2': 26, 'is2': 19, 'best2': 70}	The original dictionary 1 is : {'INDEX': 20, 'ji': 36, 'best': 100} The original dictionary 2 is : {'INDEX2': 26, 'is2': 19, 'best2': 70}
Output	Mapped dictionary : {'Gfg': 26, 'is': 19, 'best': 70}	Mapped dictionary : {'quality': 90, 'ji': 87, 'hash': 90}	Mapped dictionary : {'keys': 26, 'ji': 19, 'hash': 70}	Mapped dictionary : {'INDEX': 26, 'ji': 19, 'best': 70}